

Question			Marking details	Marks available																	
				AO1	AO2	AO3	Total	Maths	Prac												
2	(a)		Vector: Magnitude (size) and direction Scalar: Magnitude (size) only Minimum acceptable response: <i>a vector has direction</i> [1] Relevant example of each e.g. independent mark [1] <table><tr><th>Vectors</th><th>Scalars</th></tr><tr><td>Displacement</td><td>Speed</td></tr><tr><td>Velocity</td><td>Time</td></tr><tr><td>Acceleration</td><td>Distance</td></tr><tr><td>Force</td><td>Pressure</td></tr><tr><td></td><td>Temperature</td></tr></table>	Vectors	Scalars	Displacement	Speed	Velocity	Time	Acceleration	Distance	Force	Pressure		Temperature	2			2		
Vectors	Scalars																				
Displacement	Speed																				
Velocity	Time																				
Acceleration	Distance																				
Force	Pressure																				
	Temperature																				
	(b)	(i)	From graph speed = $\frac{2\,000}{100}$ or equiv [e.g. $\frac{1\,000}{50}$] (or 20 ms^{-1}) seen [1] Convincing conversion to km h^{-1} (i.e. $\times \frac{3\,600}{1\,000}$) or 72 [km h^{-1}] seen or convert 70 km h^{-1} into 19.4 m s^{-1} [1]	1	1		2	2													